

Appendix

A1 Supplementary Materials

Fig.A1 illustrates the system architecture of MAIC. Fig.A2 is a supplementary illustration of the literature surveyed in Related Works. An example of a student profile constructed by a conversational agent, which integrates academic information with personal interests extracted from dialogue, as follows.

Listing 1

```
{
  "Student Profile": {
    "Basic Information": {
      "Student ID": "student_001",
      "Profile Updated": "2025-09-12"
    },
    "Academic Profile": {
      "Year": "Freshman",
      "Major": "Business Administration"
    },
    "Interest Profile": {
      "Raw Text Input": "I love playing basketball with my dorm mates in the evenings. I usually follow NBA highlights, especially games with the Lakers. I'm also trying to improve my three-point shooting.",
      "Structured Tags (by LLM)": {
        "Domain": "Sports",
        "Category": "Basketball",
        "Sub-Category": "Playing + Watching NBA",
        "Keywords": ["Basketball", "NBA", "Lakers", "Three-Point Shooting"]
      }
    }
  }
}
```

A2 Supplementary information of Empirical Evaluation

A2.1 The quality of instruction and interaction

This study evaluates the quality of the MAIC curriculum from four main perspectives: 1) course instruction provided by AI teachers, 2) course interaction with AI teachers and peer classmates, 3) relationships with AI teachers and peer classmates, and 4) the quality of responses generated by AI agents in the system. For the first three dimensions, all students who completed the course were required to complete corresponding questionnaires ($N = 319$). Regarding instruction and interaction, the questionnaire items were adapted from the Community of Inquiry (CoI) framework to fit the characteristics of the MAIC platform and learning process [1]. Reliability and validity analyses demonstrated high overall internal consistency among all dimensions (Cronbach's $\alpha = 0.90$), with the instruction dimension at 0.87 and the interaction dimension at 0.89. For the assessment of relationships, we adapted questionnaire items from [2], with a Cronbach's α of 0.89, indicating strong reliability. These questionnaires employed a five-point Likert scale, ranging from 1 ("Strongly Disagree") to 5 ("Strongly Agree"). Furthermore, we recruited 62 students who completed the course to evaluate the responses received from the AI agent. Students assessed the responses on a 1 ("very poor") to 5 ("very good") scale across ten dimensions cat-

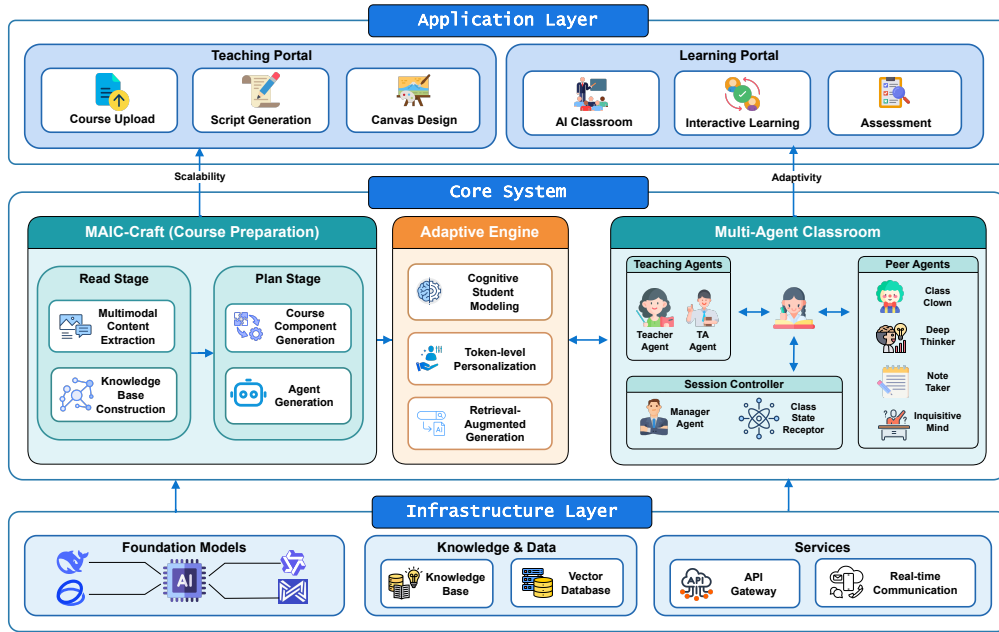


Fig.A1. Illustration of system implementation of MAIC.

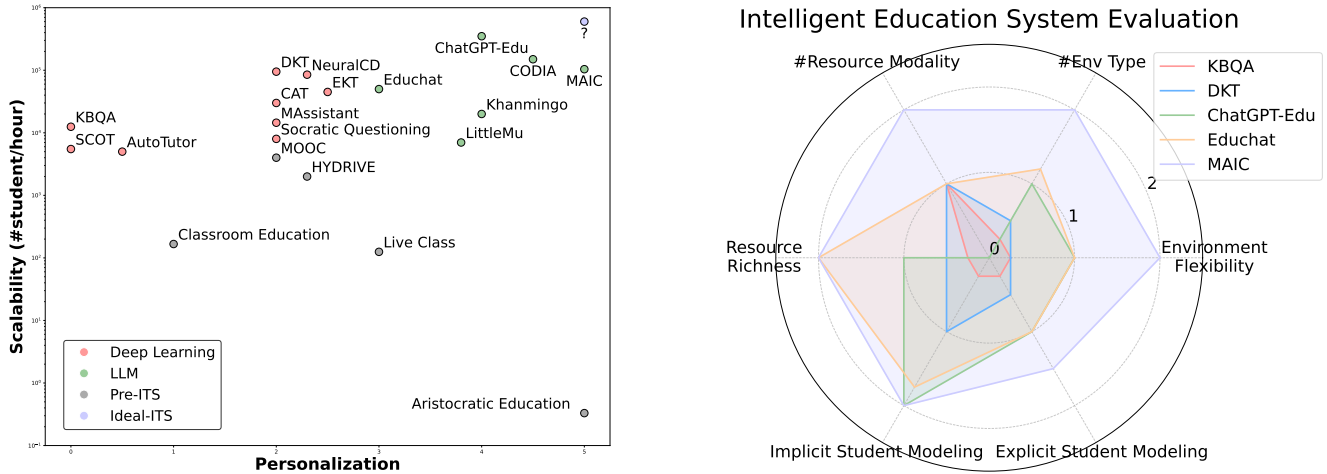


Fig.A2. A literature review of the existing education systems. The evolution trajectory illustrates a clear shift from scalability without intelligence toward adaptive AI-driven educational experiences, directing to the ideal personalized ITS at scale.

egorized into three primary areas: content quality, cognitive stimulation, and emotional support. Specifically, the ten dimensions measured were: 1) Content Quality: Content Accuracy, Content Clarity, Content Adaptability, and Content Comprehensibility; 2) Cognitive Stimulation: Logic of the Response, Open-mindedness

of the Response, and Stimulation of the Response; 3) Emotional Support: Perceived Acceptance, Perceived Enjoyment, and Perceived Support.

Fig.A3 presents the students' experience regarding the AI teacher's instruction, illustrated as a stacked bar plot. Overall, the students

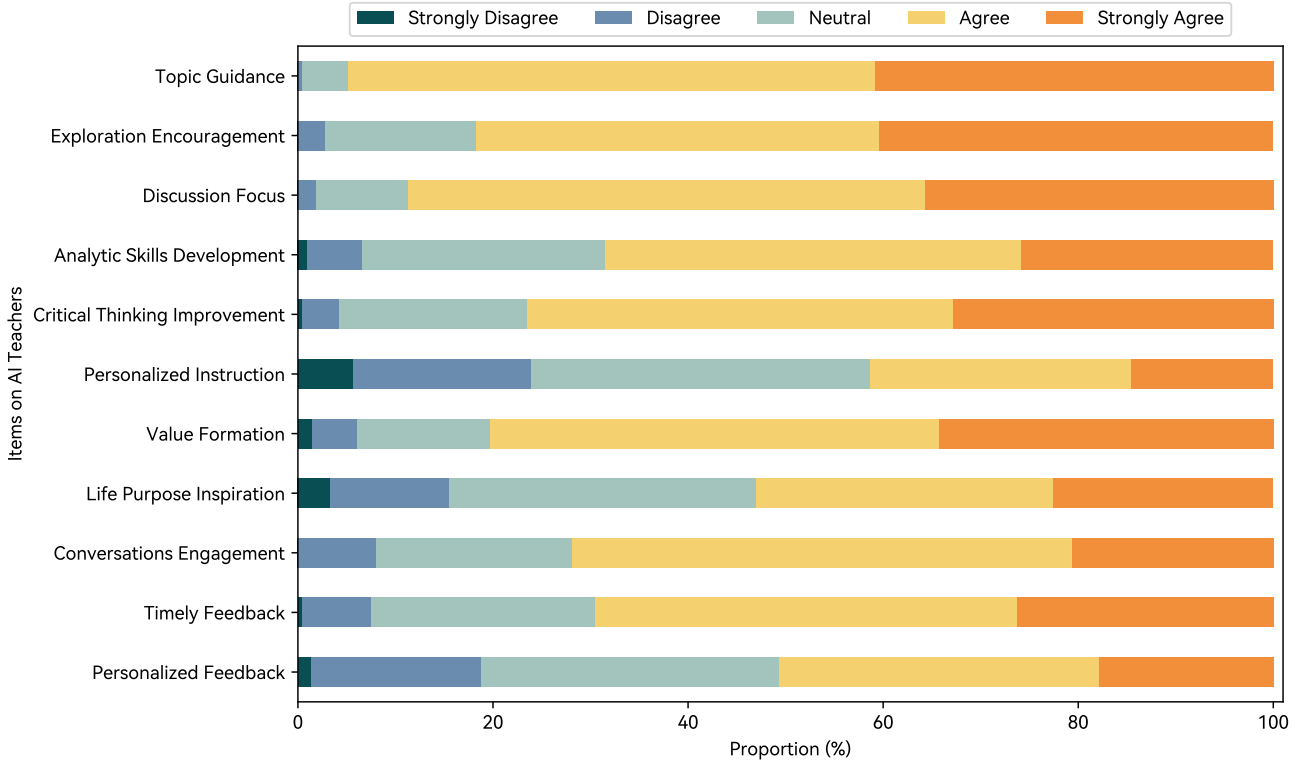


Fig.A3. Results from after-course survey on instruction.

had a positive attitude about the AI teacher's teaching practice. Specifically, the items "In this class, the teacher will guide us to understand the theme of the course, which can help me clarify my thoughts" (Topic Guidance) and "In this class, the teacher can help us focus on discussing relevant issues, which can help me learn" (Discussion Focus) received particularly strong positive feedback, with the proportion of students selecting "Agree" and "Strongly Agree" reaching 94.84% and 88.73%, respectively. These findings indicate that students widely acknowledged the AI teacher's ability to steer course direction and organize classroom discussions effectively.

However, the results indicate a necessity for

improving instructional customization, as students reported a relatively less positive experience. This might be attributed to the uniformity of the AI teacher's script, which was not tailored to individual students' learning needs. Only 41.31% of students agreed or strongly agreed that the AI teachers "pay attention to the abilities of each student and provide appropriate adjustments according to their needs" (Personalized Instruction), and 50.70% responded positively to the item "help me understand my strengths and weaknesses by providing feedback" (Personalized Feedback).

Fig.A4 displays the overall student ratings of the AI classmates. Generally, students gave positive evaluations to their AI peers. In par-

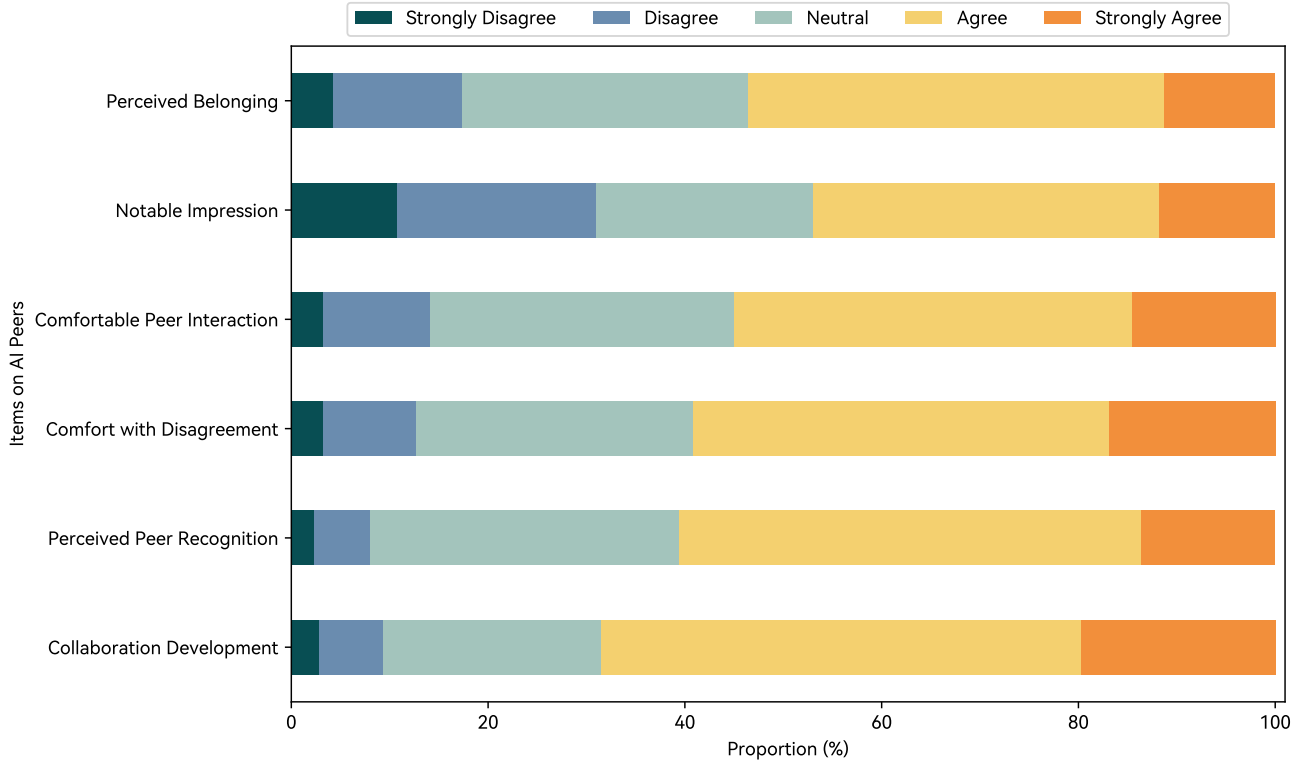


Fig.A4. Results from after-course survey on interaction.

ticular, the item "Discussing with classmates in the course can help me develop a sense of collaboration" (Collaboration Development) received relatively high ratings, with 68.54% of responses rated as "Agree" or "Strongly Agree". This result suggests that students perceived AI classmates as effective partners in fostering collaborative learning.

However, students' impressions of the AI peers were less clearer in terms of individual distinctiveness. For example, only 46.94% selected "Agree" or higher on the item "I can form clear impressions of some classmates in the course" (Notable Impression). This outcome reflects potential limitations in the current design of AI agents' personality and functionality. To

enhance students' meaningful interactions and impressions of AI classmates, it may be beneficial to reinforce features related to cognitive scaffolding or emotional support, enabling more personalized and memorable experiences within the learning environment.

A3 Ethical Considerations

The deployment of MAIC in online education brings forth significant ethical considerations that must be carefully evaluated. One of the foremost concerns is learner privacy and data security. The system's reliance on large-scale data collection and analysis to personalize learning experiences raises questions about how student data is stored, accessed, and used. To

mitigate these concerns, stringent data protection measures have been implemented, including encryption and anonymization of student records. However, given the sensitivity of educational data, continuous efforts to enhance security protocols are essential.

Issues of discrimination and bias also present ethical challenges. While MAIC aims to provide a personalized learning experience for all students, there is a risk that the algorithms driving these personalized experiences may inadvertently reinforce existing biases, particularly if the training data is not sufficiently diverse. To address this, the development team has incorporated fairness-focused auditing procedures into the algorithmic design process. Although these measures are designed to minimize bias, it is recognized that no system is entirely immune to these challenges, and ongoing monitoring is required to ensure equitable outcomes.

The accuracy of information and content regulation within MAIC is another critical area of ethical concern. As the system automates the creation and dissemination of educational content, there is a risk that inaccuracies could be propagated at scale. To mitigate this, content generated by the AI is regularly reviewed by subject matter experts and teaching assistants. Nonetheless, given the vast scale of content production, it is acknowledged that some errors may still occur. Thus, the system includes mechanisms for students and educators to flag and correct inaccuracies, ensuring continuous

improvement of the educational material.

In terms of the ethics of education, MAIC raises questions about the role of teachers in a system that increasingly relies on AI-driven instruction. While the system enhances scalability and provides personalized learning experiences, it also diminishes the direct involvement of human educators. This shift may impact the development of teacher-student relationships, which are critical for fostering emotional and social growth in learners. To address this, MAIC includes human-in-the-loop design principles, ensuring that educators can intervene and guide the AI's decision-making processes when necessary. However, the balance between AI automation and human oversight remains a complex issue that requires ongoing consideration.

Furthermore, the lack of peer interaction in a predominantly AI-driven educational environment could hinder the development of important social skills among students. To counter this, MAIC incorporates AI classmates designed to simulate peer interactions. While these agents provide a form of interaction, they cannot fully replicate the nuances of human-to-human communication. The system, therefore, encourages mixed-mode learning environments where students can engage with both AI and human peers, preserving the benefits of social learning.

Finally, the issue of student personalization must be approached with caution. While MAIC's ability to adapt to individual learning needs is a significant advantage, it also poses

risks to fairness and equality. There is a potential for certain students to receive more tailored and effective instruction based on their data profiles, potentially exacerbating educational inequalities. To mitigate this, MAIC includes mechanisms to ensure that all students, regardless of their data profiles, have access to high-quality learning experiences. The system's fairness algorithms are continuously refined to promote equitable educational outcomes for all

learners.

References

- [1] Garrison D R, Arbaugh J B. Researching the community of inquiry framework: Review, issues, and future directions. *The Internet and higher education*, 2007, 10(3):157–172.
- [2] Bastian B, Jetten J, Ferris L J. Pain as social glue: Shared pain increases cooperation. *Psychological science*, 2014, 25(11):2079-2085.